EQUALITY OF LOCAL RINGS AT A POINT WITH REDUCED FINITE-TYPE BASE FORCES A LOCAL OPEN IMMERSION

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Allowed standard prerequisites (editorial).

- Localizing ring maps and the correspondence between localizations of a Noetherian ring and affine open subschemes. Stacks Project, Algebra and Schemes (localization and affine opens). Background for replacing S and X by affine neighbourhoods and for the A_{ff'} -> B_{fg} localization argument.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes. Let $x \in X$ be a point with image $s \in S$. If*

(1) $\mathcal{O}_{X,x} = \mathcal{O}_{S,s}$,
(2) *S is reduced,*
(3) *$X \to S$ is of finite type, and*
(4) *S has finitely many irreducible components,*

then there exists an open neighbourhood U of x such that $f|_U$ is an open immersion.

Proof. We may remove the (finitely many) irreducible components of S which do not contain s. We may replace S by an affine open neighbourhood of s. We may replace X by an affine open neighbourhood of x. Say $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$. Let $\mathfrak{q} \subset B$, resp. $\mathfrak{p} \subset A$ be the prime ideal corresponding to x, resp. s. As A is a reduced and all of the minimal primes of A are contained in $\mathfrak{p}$ we see that $A \subset A_\mathfrak{p}$. As $X \to S$ is of finite type, B is of finite type over A. Let $b_1, \ldots, b_n \in B$ be elements which generate B over A Since $A_\mathfrak{p} = B_\mathfrak{q}$ we can find $f \in A$, $f \notin \mathfrak{p}$ and $a_i \in A$ such that b_i and a_i/f have the same image in $B_\mathfrak{q}$. Thus we can find $g \in B$, $g \notin \mathfrak{q}$ such that $g(fb_i - a_i) = 0$ in B. It follows that the image of $A_f \to B_{fg}$ contains the images of $b_1, \ldots, b_n$, in particular also the image of g. Choose $n \geq 0$ and $f' \in A$ such that f'/f^n maps to the image of g in B_{fg}. Since $A_\mathfrak{p} = B_\mathfrak{q}$ we see that $f' \notin \mathfrak{p}$. We conclude that $A_{ff'} \to B_{fg}$ is surjective. Finally, as $A_{ff'} \subset A_\mathfrak{p} = B_\mathfrak{q}$ (see above) the map $A_{ff'} \to B_{fg}$ is injective, hence an isomorphism. □

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